WEEK 2

PROGRAM 1

```
#include <iostream>
using namespace std;
void swap_by_reference(int &a, int &b) {
    int temp;
    temp = a;
    a = b;
    b = temp;
}
int main() {
    int x = 10, y = 20;
    cout << "Before swap: x = " << x << ", y = " << y << endl;
    swap_by_reference(x, y);
    cout << "After swap: x = " << x << ", y = " << y << endl;
    return 0;
}
```

PROGRAM 2

```
#include <iostream>
using namespace std;
void swap_by_value(int a, int b) {
```

```
    int temp;
    temp = a;
    a = b;
    b = temp;
    cout << "Inside function: a = " << a << ", b = " << b << endl;
}
int main() {
    int x = 10, y = 20;
    cout << "Before function call: x = " << x << ", y = " << y << endl;
    swap_by_value(x, y);
    cout << "After function call: x = " << x << ", y = " << y << endl;
    return 0;
}
```

PROGRAM 3

```
#include <iostream>
using namespace std;
int main() {
    int n;
cout << "Enter the number of elements (n): ";
cin >> n;
    int* arr = new int[n];
```

```
    cout << "Enter " << n << " integers:" << endl;
    for (int i = 0; i < n; ++i) {
        cout << "Element " << i + 1 << ": ";
       cin >> arr[i];
    }
   cout << "\nYou entered the following elements:" << endl;
    for (int i = 0; i < n; ++i) {
       cout << "Element " << i + 1 << ": " << arr[i] << endl;
    }


    return 0;
}
```

PROGRAM 4

```
#include <iostream>
using namespace std;
int main() {
    int data[5];
    int* ptr;
    ptr = data;
    cout << "Enter 5 elements:" << endl;
    for (int i = 0; i < 5; ++i) {
```

```
        cin >> *(ptr + i);
    }
    cout << "You entered:" << endl;
    for (int i = 0; i < 5; ++i) {
        cout << *(data + i) << endl;
    }
    return 0;
}
```

PROGRAM 5

```
#include <iostream>
using namespace std;
int main()
{
    int n, i;
    int *p;
    cout << "Enter n value: ";
    cin >> n;
    p = new int[n];
    cout << "Enter array elements: ";
    for(i = 0; i < n; i++)
    {
```

```
        cin >> *(p + i);
    }
    cout << "Array elements are: ";
    for(i = 0; i < n; i++)
    {
        cout << *(p + i) << " ";
    }
    delete[] p;
    return 0;
}
```

PROGRAM 6

```
#include <iostream>
#include <string>
using namespace std;
void display(int a) {
    cout << "Displaying one integer: " << a << endl;
}
void display(int a, int b) {
    cout << "Displaying two integers: " << a << ", " << b << endl;
}
void display(double a) {
```

```
    cout << "Displaying one double: " << a << endl;
}
void display(string a) {
    cout << "Displaying one string: " << a << endl;
}
void display_order(int a, string b) {
    cout << "Order (int, string): " << a << ", " << b << endl;
}
void display_order(string a, int b) {
    cout << "Order (string, int): " << a << ", " << b <<endl;
}
int main() {
    display(10);
    display(10, 20);
    display(12.34);
    display("Hello");
    display_order(5, "world");
    display_order("world", 5);
    return 0;
}
```